CATCH: Counterfactual Anatomical Tissue Inpainting with Conditional Haar Diffusion

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Abstract. The BraTS local-synthesis task requires plausible tumor-free tissue to be generated inside a masked region of a T1-weighted brain MRI while preserving the observed anatomy. We present CATCH, a conditional 3D diffusion framework operating on an invertible Haar-wavelet representation. The denoiser receives noisy target coefficients, voided-image coefficients, and a signed mask. Training combines tumor-excluded wavelet reconstruction with a hole-focused loss, while hard compositing preserves the observed image. We compare fixed-mask training, tumor-component random augmentation, and a weighted mixture of tumor-derived, irregular-blob, and ellipsoidal masks. Checkpoints and trajectory-aggregation policies are selected on development data and frozen for evaluation on a held-out 75-case confirmation set restricted to artificially removed healthy tissue. The selected weighted-mixture pipeline, averaging five stochastic trajectories, achieved SSIM 0.80 ± 0.13, PSNR 19.18 ± 1.80 dB, and MSE 0.010 ± 0.005 (mean±SD). Relative to compute-matched random augmentation, it improved SSIM by 0.019 (95% bootstrap CI: 0.013–0.025), PSNR by 0.95 dB, and MSE by 0.003; all three paired comparisons remained significant after Holm correction. Fixed-mask single-trajectory inference achieved SSIM 0.76 ± 0.16. These results favor the complete weighted-mixture policy within CATCH, but the absence of an external baseline limits broader conclusions.

Keywords: Brain MRI inpainting · Healthy-tissue synthesis · Denoising diffusion models · Wavelet transform · Medical image synthesis · BraTS

1 Introduction

Brain tumors alter local anatomy and can undermine registration, parcellation, tissue-segmentation, and atlas-based pipelines developed for non-pathological brains [1]. Healthy-tissue inpainting replaces a pathological region with plausible tumor-free anatomy so that downstream tools can operate on a completed image. The BraTS local-synthesis task [2] formalizes this setting: given a voided T1-native (T1n) volume and an inpainting mask, synthesize the missing anatomy while leaving the observed image unchanged.

Denoising diffusion probabilistic models (DDPMs) [3] provide a flexible framework for conditional image generation [4,5], but direct full-resolution 3D diffusion is computationally demanding. Wavelet diffusion models (WDMs) [6] reduce the spatial cost via a fixed invertible transform, and conditional WDM [7] extends this idea to paired volumetric synthesis. We adapt this framework as CATCH and study how artificial healthy-tissue holes should be sampled during training.

Training only on supplied masks offers limited spatial and morphological variation. Additional holes may improve coverage, but their source, size, shape, placement, and refresh strategy can alter the learned reconstruction distribution. We therefore compare three mask policies within the same conditional 3D wavelet-diffusion framework. Our main contributions are: (1) a full-volume conditional 3D Haar-wavelet diffusion model with tumor-excluded supervision and hard preservation of observed voxels; (2) a controlled comparison of fixed masks, tumor-component augmentation, and a weighted mixture of tumor-derived, irregular blob, and ellipsoidal masks, including a compute-matched comparison of the two augmented pipelines; and (3) development-only model and inference selection followed by frozen 75-case confirmation with paired uncertainty, statistical testing, patient-disjoint sensitivity analysis, and qualitative inspection.

2 Related Work

Diffusion and inpainting. DDPMs learn the reverse of a gradual noising process [3], while score-based formulations connect diffusion to stochastic differential equations [8]. RePaint conditions an unconditional model by repeatedly restoring known pixels during sampling [4], whereas Palette directly supplies the observed image to a conditional denoiser [5]. CATCH follows the latter strategy and applies final hard compositing.

Volumetric medical synthesis. Latent diffusion reduces high-resolution cost through a learned latent space [9]. WDM instead applies an exactly invertible 3D wavelet transform [6], and its conditional extension supports volumetric image-to-image synthesis [7]. Diffusion has also been used for 3D healthy-brain inpainting [10] in the BraTS local-synthesis setting [2], based on the broader BraTS benchmark [11,12,13]. fastWDM3D further reduces wavelet-domain sampling cost through a shortened reverse process [14]. Our work retains full ancestral sampling and focuses on mask representation, tumor-excluded supervision, online healthy-hole sampling, and trajectory aggregation. Differences in data and evaluation preclude direct numerical comparison with prior scores.

3 Method

CATCH adapts Tooth-Diffusion [15] from dental CBCT synthesis to brain-MRI inpainting. Let x_v denote the voided T1n volume and $m \in \{0,1\}^{H\times W\times D}$ the complete inpainting region. During training, m is decomposed into disjoint healthy and pathological components, $m = m_h \vee m_u$ and $m_h \odot m_u = 0$. The complete T1n image x provides a valid target in m_h but remains pathological in m_u, which is therefore excluded from supervision. At inference, only x_v and m are available to the model.

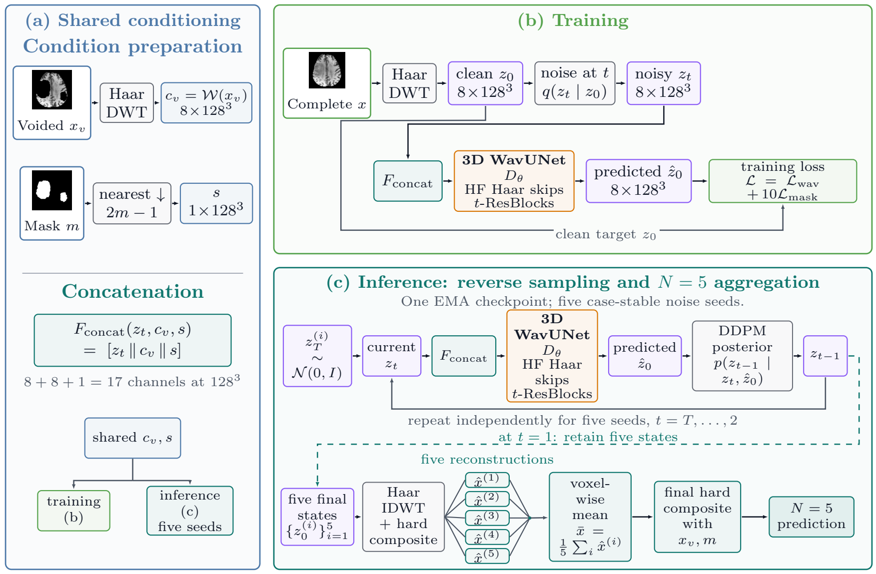

Fig. 1: CATCH training and inference. A one-level 3D Haar transform produces eight subbands. Noisy target coefficients, voided-image coefficients, and the signed mask form the 17-channel WavUNet input. Predictions are inverse-transformed and hard-composited with the observed image. Random and weighted-mixture pipelines average five case-stable noise trajectories from one exponential-moving-average (EMA) checkpoint; fixed mask uses $N = 1$.

3.1 Conditional Wavelet Diffusion

Volumes are reoriented to RAS and center-padded from $240 \times 240 \times 155$ to 256^3 without resampling. A one-level 3D Haar transform $\mathcal{W}$ produces eight 128^3 subbands. Following WDM [6], the LLL channel is divided by 3 after the forward transform and restored before inversion for target and conditioning coefficients. The transform is therefore fixed and invertible rather than a learned autoencoder.

Let $z_0 = \mathcal{W}(x)$ denote the clean target coefficients. With $T = 1000$, $\alpha_t = 1 - \beta_t$, and $\bar{\alpha}_t = \prod_{j=1}^{t} \alpha_j$, the forward process is

$$q(z_t \mid z_0) = \mathcal{N}\big(z_t; \sqrt{\bar{\alpha}_t}\, z_0, (1 - \bar{\alpha}_t)\mathbf{I}\big)\,, \tag{1}$$

where $\{\beta_t\}_{t=1}^{T}$ discretizes the variance-preserving SDE schedule [8] using the continuous rate parameters $\beta_{\min} = 0.1$ and $\beta_{\max} = 20$. The network predicts $\hat{z}_0$ directly; ancestral sampling uses the DDPM posterior parameterized by $(z_t, \hat{z}_0)$ with the `ModelVarType.FIXED_LARGE` reverse-variance option.

The voided image is transformed as $c_v = \mathcal{W}(x_v)$. The mask is nearest-neighbor downsampled and encoded as $s = 2\,\mathrm{NN}(m) - 1 \in \{-1, 1\}^{128^3}$. The denoiser therefore receives

$$\hat{z}_0 = D_\theta([z_t \| c_v \| s], t), \tag{2}$$

a 17-channel input. D_θ is a fully 3D wavelet U-Net [16,17,6], referred to as *WavUNet* here, with base width 64, channel multipliers $(1,2,2,4,4,4)$, two residual blocks per level, GroupNorm, SiLU, timestep-conditioned scale-and-shift modulation, wavelet skip connections, and no self-attention.

3.2 Healthy-Tissue Objective

Target and conditioning image share the inference-available scale $a = \max(x_v)$. Intensities are divided by a, clipped to $[0,1]$, and mapped to $[-1,1]$. Let $e_{k,p} = (\hat{z}_0^{(k)}(p) - z_0^{(k)}(p))^2$ denote the error in subband k at wavelet location $p \in \Omega$. We set $u_p = 0$ if the corresponding 2^3 voxel cell intersects m_u, and $u_p = 1$ otherwise:

$$\mathcal{L}_{\text{wav}} = \frac{1}{8|\Omega|} \sum_{k=1}^{8} \sum_{p \in \Omega} u_p e_{k,p}. \tag{3}$$

This matches the implemented fixed-grid normalization: cells touching the real tumor contribute zero error but remain part of the fixed denominator.

For the hole-focused term, cells intersecting m_h form a binary coverage map that is Gaussian-smoothed ($\sigma = 2$ on the wavelet grid) and multiplied by u to obtain $w_p \geq 0$:

$$\mathcal{L}_{\text{mask}} = \frac{\sum_{k=1}^{8} \sum_{p \in \Omega} w_p e_{k,p}}{8 \sum_{p \in \Omega} w_p}, \quad \mathcal{L} = \mathcal{L}_{\text{wav}} + 10\mathcal{L}_{\text{mask}}. \tag{4}$$

When $\sum_p w_p > 0$, the implementation uses Eq. (4); if this support is zero, it assigns that example's regional term zero rather than adding an epsilon. Both terms exclude cells touching the real tumor.

3.3 Healthy-Mask Training Variants

The *fixed-mask* variant uses the supplied m_h. Online variants sample m_h', void $m_h' \cup m_u$, and supervise m_h' while excluding m_u. Their tumor-component bank contains only the 1,151 optimization cases, with five repeat slots per case.

Random augmentation. This reference-style policy uses tumor components only. Current-case components are allowed; source masks are sampled by inverse tumor-to-brain-ratio rank within a conditioned 10% window, transformed by the reference flips and rotations, and fixed across epoch runs. Exhausted placement raises an error rather than silently changing the policy.

Weighted-mixture augmentation. This policy samples 80% tumor components, 15% irregular blobs, and 5% ellipsoids. Current-case tumor components are excluded; empirical target volumes, bounded 3D rotations, optional anisotropic scaling, and epoch-wise slot refresh are used. Accepted masks are connected, contain 800–225,070 voxels, remain at least five voxels from m_u, and overlap background by at most 25%. Exhausted generation falls back to a nonempty supplied healthy mask that does not overlap m_u.

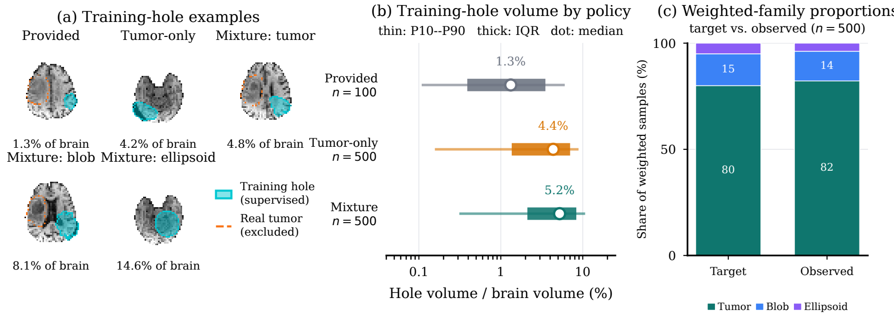

Fig. 2: Training-mask audit. (a) Supplied and generated masks; cyan and orange denote supervised voxels and excluded tumor. (b) Epoch-0 hole-volume distributions (10th–90th percentile, interquartile range, and median; $n = 100/500/500$). (c) Target and observed weighted-mixture proportions: tumor/blob/ellipsoid 80/15/5% and 82.2/14.0/3.8% (411/70/19); no fallback occurred.

3.4 Sampling and Compositing

We use exponential-moving-average (EMA) weights and the full 1000-step ancestral reverse process. At each step, `clip_denoised=True` inverse-transforms the predicted clean state, clips it in normalized voxel space, and returns it to the wavelet domain. The final reconstruction is restored to native scale, shape, and orientation and hard-composited:

$$\hat{x} = x_v \odot (1 - m) + \hat{x}_{\text{gen}} \odot m. \tag{5}$$

So, outside the hole, the prediction is explicitly copied from the voided input.

Global seed 0 and the case- and member-stable `brats-validation-noise-v1` scheme make trajectories shard-independent and preserve the first members across larger N. Multi-trajectory policies independently decode and composite each sample, aggregate volumes voxel-wise, and composite once more. Thus, trajectories come from one checkpoint rather than independently trained models.

4 Experiments

4.1 Data and Split Protocol

The ASNR-MICCAI BraTS Local Synthesis release [2,11,12,13,18,19,20] contains 1,251 training cases with complete and voided T1n volumes, a complete inpainting mask, and healthy and unhealthy submasks. However, the organizer-validation ground truth is hidden.

A deterministic 100-case hold-out (seed 2026) leaves 1,151 optimization cases, excluded from the hold-out loader and component bank. The hold-out is split into 25 development and 75 confirmation cases. Development and confirmation patients are disjoint, but the split is case-keyed: 3/25 development and 11/75 confirmation cases have sibling timepoints in the pool. None of the five checkpoint-diagnostic cases has such a sibling. We therefore also report a post-hoc analysis on the 64 confirmation cases that are patient-disjoint from optimization.

4.2 Training Configuration

The three 17-channel models differ only in healthy-mask policy. Each uses seed 0, batch size one per GPU, AdamW [21] with learning rate 10^{-5} and zero weight decay, linear decay over a 300,000-step horizon, and EMA decay 0.9999. Training is FP32 on NVIDIA A100 40 GB GPUs. Periodic checkpoints are saved every 5,000 steps through 295,000, followed by a separate terminal save at step 299,999 that includes the EMA state.

4.3 Checkpoint and Inference-Policy Selection

Selection uses only the 25-case development set and consists of two stages.

Checkpoint selection. Every 10,000 steps, one deterministic 1000-step trajectory is evaluated on five fixed development cases; fixed-mask step 200,000 is unavailable. Within each training arm, checkpoints are ranked by aggregate SSIM, PSNR, and MSE, and the lowest mean rank is retained. Fixed-mask steps 170,000 and 180,000 tie; step 180,000 is chosen because it leads in PSNR and MSE. Random-augmentation step 290,000 and weighted-mixture step 200,000 uniquely minimize mean rank. These five-case diagnostics are used only for checkpoint selection and are not reported as final performance estimates.

Trajectory-policy selection. At each terminal EMA state, all 25 development cases compare nested $N \in \{1, 3, 5\}$ trajectories with voxel-wise mean or median aggregation. Mean $N = 5$ is selected for both augmented arms. For fixed mask, mean $N = 3$ lowers joint rank from 2.920 to 2.787 but triples inference cost, while mean $N = 5$ is worse (3.240); we therefore retain $N = 1$. Each selected policy is then transferred to the corresponding selected checkpoint and frozen before confirmation. This design reduces development-time computation but assumes that the preferred aggregation policy is sufficiently stable across late checkpoints.

4.4 Evaluation and Statistical Analysis

The official evaluator normalizes prediction and target to $[0, 1]$ using the voided-image 0.5th and 99.5th percentiles and scores only the supplied healthy submask. We report SSIM, PSNR, and MSE. Means and sample standard deviations describe case heterogeneity, not training-seed variability.

For case i and pipeline j, let r_{ij}^{SSIM}, r_{ij}^{PSNR}, and r_{ij}^{MSE} denote ranks among the selected pipelines. Mean joint rank is

$$R_j = \frac{1}{3n} \sum_{i=1}^{n} \left(r_{ij}^{\mathrm{SSIM}} + r_{ij}^{\mathrm{PSNR}} + r_{ij}^{\mathrm{MSE}} \right), \tag{6}$$

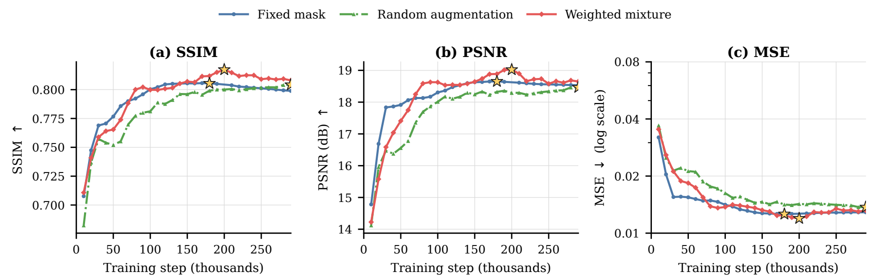

Fig. 3: Five-case EMA checkpoint diagnostics. Stars mark retained checkpoints after ranking aggregate SSIM, PSNR, and MSE. Fixed-mask steps 170k and 180k tie in mean rank; step 180k is retained because it leads in PSNR and MSE. Each point uses one deterministic 1000-step trajectory; curves are selection diagnostics, not confirmation results.

with $n = 75$ for confirmation; development configurations are ranked analogously within each training arm.

For each metric, a Friedman test assesses the omnibus difference and two-sided paired Wilcoxon tests assess the three pairwise contrasts, with Holm correction applied within metric. The SciPy calls set `zero_method` to `wilcox`, force the asymptotic approximation, and apply no continuity correction; none of the reported paired differences is zero. Mean paired differences use deterministic 100,000-resample bootstrap 95% CIs (seed 2026). Predictions, per-case metrics, split audits, hashes, and run metadata are retained in CSV/JSON format. Code and scripts are available at `https://github.com/simonwinther/brats2026`.

5 Results

5.1 Development Selection

Figure 3 shows the 5-case diagnostics and selected EMA checkpoints. Fixed-mask steps 170,000 and 180,000 tie for the lowest mean rank; step 180,000 is retained because it leads in PSNR and MSE. Random step 290,000 and weighted-mixture step 200,000 are best on all aggregate metrics within their respective histories.

Table 1 summarizes trajectory-policy selection on all 25 development cases. Mean $N = 5$ has the lowest joint rank for both augmented models. For fixed mask, mean $N = 3$ improves rank by only 0.133 over $N = 1$ at triple trajectory cost, and mean $N = 5$ is worse. We therefore select fixed mask with $N = 1$ and both augmented models with mean $N = 5$.

5.2 Case-Disjoint 75-Case Confirmation

Table 2 summarizes the frozen confirmation results. Omnibus differences are significant for SSIM ($\chi^2_F(2) = 52.83$), PSNR, and MSE (both $\chi^2_F(2) = 52.91$; all

Table 1: Development-set joint rank at the terminal EMA states. Bold and underlining mark the lowest and second-lowest values within each arm. Fixed-mask $N = 1$ was retained for the compute–performance trade-off.

Configuration	Fixed mask	Random aug.	Weighted mixture
$N = 1$	<u>2.920</u>	4.707	4.400
Mean $N = 3$	**2.787**	<u>2.133</u>	2.693
Median $N = 3$	2.973	3.533	3.773
Mean $N = 5$	3.240	**1.600**	**1.587**
Median $N = 5$	3.080	3.027	<u>2.547</u>

Table 2: Case-disjoint 75-case confirmation results (mean ± SD). Lower joint rank is better. Bold and underlining indicate the best and second-best results, respectively. The fixed-mask pipeline uses $N = 1$ and is therefore not compute-matched to the two mean-$N = 5$ pipelines.

Pipeline	SSIM ↑	PSNR (dB) ↑	MSE ↓ ($\times 10^{-2}$)	Rank ↓
Fixed ($N = 1$)	0.756±0.162	17.84±1.75	1.365±0.807	2.551
Random (mean $N = 5$)	<u>0.778±0.140</u>	<u>18.24±2.19</u>	<u>1.252±0.718</u>	<u>2.076</u>
Weighted (mean $N = 5$)	**0.797±0.131**	**19.18±1.80**	**0.987±0.540**	**1.373**

$p < 3.4 \times 10^{-12}$). Relative to fixed mask, weighted mixture improves SSIM by 0.041 (95% CI: 0.032–0.050), PSNR by 1.34 dB (1.08–1.61), and $[0, 1]$-scale MSE by 0.0038 (0.0029–0.0047), although this comparison is not compute-matched.

Relative to compute-matched random augmentation, weighted mixture improves SSIM by 0.019 (0.013–0.025), PSNR by 0.95 dB (0.66–1.27), and $[0, 1]$-scale MSE by 0.0026 (0.0019–0.0034). All three pairwise tests within each metric remain significant after Holm correction; the largest adjusted p-value is 0.0178.

Figure 4 shows that the compute-matched gain occurs in most cases. Weighted mixture outperforms random augmentation in 54/75 cases for SSIM and 57/75 for PSNR and MSE. It improves all three metrics in 51 cases, loses all three in 15, and has mixed effects in 9. Across all pipelines, it is simultaneously best on all three metrics in 46 cases and attains the lowest mean joint rank.

Figure 5 shows variation in case difficulty. The low-SSIM example preserves gross structure but shows localized intensity and boundary mismatch; the median retains residual error; and the high-SSIM example closely matches the observed target. Because the observed T1n may remain pathological elsewhere in the full inpainting region, these scores do not establish counterfactual accuracy there.

6 Discussion

Weighted mixture is the strongest pipeline on the confirmation set. Random and weighted mixture use the same architecture, training horizon, mean-$N = 5$

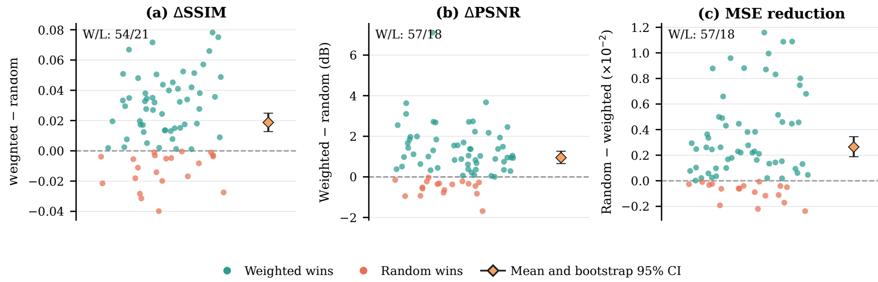

Fig. 4: Compute-matched paired effects of weighted mixture versus random augmentation across 75 cases. Positive values favor weighted mixture. Points represent cases; diamonds show mean paired effects with 100,000-resample case-bootstrap 95% CIs. W/L denotes wins/losses.

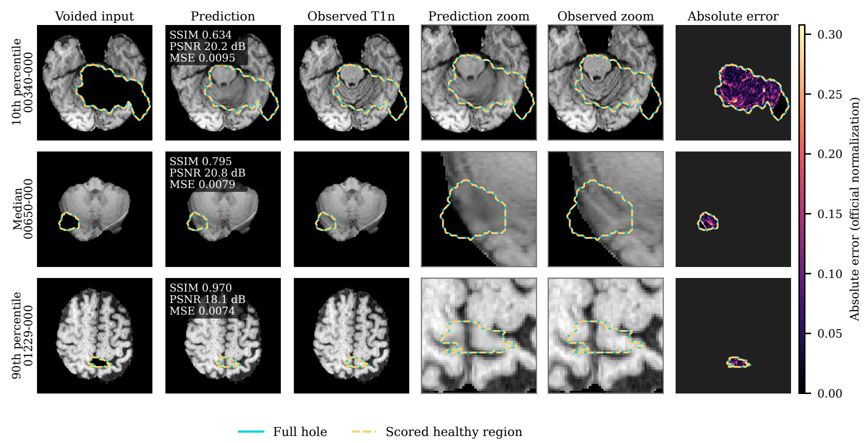

Fig. 5: Weighted-mixture mean-$N = 5$ reconstructions at the empirical 10th, 50th, and 90th SSIM percentiles. Cyan denotes the complete inpainting region and dashed yellow the scored healthy region. Error is shown only within the scored region on a shared clipped scale.

inference, split, noise trajectories, and evaluator, so their comparison is compute-matched. Their training policies nevertheless differ jointly in mask families, volume sampling, transforms, current-case eligibility, epoch refresh, and fallback. The evidence therefore supports the complete policy rather than its 80/15/5 proportions or any single component. Improvements in most cases argue against an outlier-only effect, but 15 all-metric losses show that the method is not uniformly superior. One training seed and arm-specific checkpoints limit attribution.

The fixed-mask pipeline uses one trajectory. Fixed mask uses $N = 1$ because mean $N = 3$ provided only a small development-rank gain at triple cost and mean

$N = 5$ was worse. Comparisons with fixed mask therefore combine training-policy and inference-compute differences.

Wavelets enable full-volume 3D sampling, while hard compositing preserves the observed input outside the hole. However, mean-$N = 5$ inference requires five 1000-step trajectories per case. Faster wavelet inpainting [14] should therefore be evaluated under the same cohort and protocol.

Evaluation covers only artificially removed healthy voxels. Since the observed T1n remains pathological in m_u, neither quantitative metrics nor qualitative references establish patient-specific counterfactual correctness in the real tumor region. Plausible reconstructions may still hallucinate incorrect anatomy, and clinical validity remains untested. Hidden organizer-validation ground truth also prevents local external evaluation.

The split is case-keyed rather than fully patient independent: the optimization pool contains sibling timepoints for 3/25 development and 11/75 confirmation cases. In the 64 confirmation cases patient-disjoint from optimization, weighted mixture obtains SSIM 0.810, PSNR 19.27 dB, and MSE 0.0092; random obtains 0.791, 18.33 dB, and 0.0117; and fixed mask obtains 0.770, 17.93 dB, and 0.0128. The ordering is unchanged, although development-stage selection may still be influenced by optimization-pool siblings.

Other limitations are one seed per arm, five-case single-trajectory checkpoint selection, transfer of terminally selected inference policies to arm-specific checkpoints, and no external baseline. Across-case SDs and bootstrap CIs do not measure training-seed variability, so superiority to prior methods is unestablished. Development predictions and one random-arm worktree difference were not fully preserved, although configurations, metrics, metadata, prediction hashes, and final confirmation outputs remain archived. Future work should add seeds and baselines, evaluate faster samplers, adopt patient-level splits, and investigate multimodal conditioning and anatomy-aware plausibility.

7 Conclusion

We presented CATCH, a conditional 3D Haar-wavelet diffusion framework for healthy brain-tissue inpainting. The method combines full-volume wavelet-domain diffusion, tumor-excluded reconstruction supervision, hole-focused loss weighting, and hard compositing of the observed image. Within CATCH, weighted-mixture augmentation outperformed compute-matched tumor-component augmentation on the 75-case confirmation set, with the same ordering in the patient-disjoint sensitivity analysis. These findings support the complete weighted-mixture policy under the reported configuration, but the lack of an external baseline and evaluation only on artificially removed healthy tissue limit broader claims.

Acknowledgments. This project is supported by the Pioneer Centre for AI, funded by the Danish National Research Foundation (grant number P1). Data used in this publication were obtained as part of the Challenge project through Synapse ID (syn74274097).

Disclosure of Interests. The authors declare no competing interests.

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